

Ashfaque Khowaja

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CAREER PROFILE

Dedicated PhD student at the University of New South Wales with a robust background in Computer Science and Engineering related to Medical Image processing, Computer Vision and Data Mining
Highly adaptable professional with proven communication, team management, complaint handling, and organisational skills
Seeking Intern opportunities to contribute technical expertise and collaborative spirit while pursuing advanced research related to Machine Learning

EDUCATION

PhD in Computer Science and Engineering February 2026
University of New South Wales

- Engaging in research on Medical Image Processing using Deep Learning, especially in MRI and CT, focusing on dementia and Alzheimer's Disease.

Master of Engineering in Computer Science and Technology September 2021 - June 2024
Central South University, China

- Awarded University Scholarship (2021-2024)
- Won 1st position in 21-day clock-in activity (2022)
- Thesis: Computer Vision and Data Mining Approaches for Medical Image Analysis (Cervical Cancer Detection in Pap-Smear Images)

Bachelor of Engineering in Computer Science and Technology September 2017 - June 2021
Guilin University of Electronic Technology, China

- Outstanding Graduate Award (2021)
- Merit Partial Tuition Fee Scholarship (2018-2021)

RELEVANT EXPERIENCE

Research Assistant September 2024 - December 2025
Southern University of Science and Technology (SUSTech), China

- Collaborated with interdisciplinary research groups and industry partners on machine learning and computer vision projects
- Led the Electronics and Hardware teams to finish projects before deadlines and make sure everything is ready to be delivered
- Managed research reports, work reports, and technical guiding documents
- Utilised KiCad software to design circuit boards, soldering, and troubleshooting
- Built embedded hardware devices, combining hardware and software

RELEVANT PROJECTS

Medical Image Analysis for Cervical Cancer Detection (Master's Research) September 2021 - June 2024
Central South University, China

- Developed deep learning models (CNNs, Vision Transformers) for automated cervical cancer classification in Pap-smear images using computer vision and data mining techniques
- Implemented attention mechanisms and weakly supervised learning to improve diagnostic accuracy and reduce false positives
- Performed large-scale data preprocessing, feature extraction, and pattern recognition on medical imaging datasets

- Achieved state-of-the-art performance metrics; results published in peer-reviewed journals and conferences (IF: 4.2)
- Technologies: Python, TensorFlow/PyTorch, OpenCV, Scikit-learn, Data Mining, Computer Vision, Pattern Recognition

Electrical Impedance Tomography (EIT) Research

February 2025 - December 2025

SUSTech, China

- Designed and simulated analogue and digital circuits for Electrical Impedance Tomography applications, focusing on signal acquisition and accuracy optimisation
- Conducted experimental testing, data analysis, and literature review to improve imaging performance and reduce noise interference
- Collaborated with supervisors to evaluate system performance and propose enhancements for biomedical applications
- Technologies: Python, Electronics, PCB Circuit Design

Dynamic Pin-Array Table

August 2025 - December 2025

SUSTech, China

- Developed a dynamic pin-array system capable of real-time surface reconfiguration using an embedded control mechanism
- Implemented hardware-software integration to optimise actuation control and improve system responsiveness
- Technologies: Embedded Systems, Python, Real-Time Control, Hardware-Software Integration

Eye Massager Electronic Design

October 2024 - March 2025

SUSTech, China

- Designed, simulated, and tested electronic circuits for a wearable eye massager, focusing on safe voltage regulation and efficient power management
- Integrated microcontroller-based control for vibration and heat functions to enhance user comfort and device performance
- Performed hardware debugging and validation to ensure reliability and compliance with wearable health device standards
- Technologies: PCB Circuit Design, Microcontroller Programming, Power Management

PUBLICATIONS

- Khowaja, A., Zou, B., & Kui, X.Y. Enhancing Cervical Cancer Diagnosis: Integrated Attention-Transformer System with Weakly Supervised Learning. Image and Vision Computing. IF: 4.2
- Khowaja, A., Zou, B., & Kui, X.Y. CerviNet A Novel Approach for Pap-smear Images Analysis for Cervical Cancer Classification. International Symposium on Bioinformatics Research and Applications (ISBRA)
- Talpur, D., Ali Raza., Khowaja, A., DeepCervixNet: An Advanced Deep Learning Approach for Cervical Cancer Classification in Pap Smear Images. VAWKUM Transactions on Computer Sciences, 2024; 12(1), 136-148
- Khowaja, A., Zou, B., & Kui, X.Y. Cervix Visionator ELM: A Novel Approach to Early Detection of Cervical Cancer. 8th International Conference on Control, Robotics, and Cybernetics (CRC 2023)

EXTRACURRICULAR ACTIVITIES

- Subcommittee Member (2026): UNSW Postgraduate Council for International Students
- Ambassador, (2024): International Students at Central South University
- Mentor, Residential Camp (2023): Provided guidance and support to camp participants
- Member, Youth Parliament (2022): Engaged in leadership and civic activities
- Member, International Student Union, GUET (2019): Promoted mutual growth and community engagement
- Vlogging & Content Creation: Awarded Participation Certificate in vlogging competition (2019)

TECHNICAL SKILLS

- Machine Learning & AI: Deep Learning, Computer Vision, Pattern Recognition, Data Mining, CNNs, Vision Transformers, Attention Mechanisms, Weakly Supervised Learning, Classification Algorithms, Model Optimisation, Feature Engineering
- Programming & Frameworks: Python, SQL, MATLAB, R, TensorFlow, PyTorch, Keras, OpenCV, Scikit-learn, Pandas, NumPy
- Data & Analytics: Data Preprocessing, Statistical Analysis, Large-Scale Data Analysis, Experimental Design
- Tools & Software: KiCad (PCB Design), LaTeX, Git, Jupyter Notebook, Google Colab, Microsoft Office, Google Docs
- Hardware & Embedded: Embedded Systems, Circuit Design, Microcontroller Programming, Hardware-Software Integration, Signal Processing

PROFESSIONAL SKILLS

- Languages: English (Proficient), Chinese (HSK 4 - Proficient), Urdu/Sindhi (Native), Hindi (Speaking)
- Soft Skills: Cross-functional Collaboration, Written and Verbal Communication, Technical Writing, Team Leadership, Project Management, Problem-Solving